STAD29 / STA 1007 assignment 2

Due Tuesday Jan 29 at 11:59pm on Blackboard

Hand in the indicated questions. In preparation for the questions you hand in, it is worth your while to work through (or at least read through) the other questions as well.

Hand in your work on Quercus. If you did STAC32 last fall, it's the same procedure. A reminder is here: `https://www.utsc.utoronto.ca/~butler/c32/quercus1.nb.html`

You are reminded that work handed in with your name on it must be *entirely your own work*. It is as if you have signed your name under it. If it was done wholly or partly by someone else, *you have committed an academic offence*, and you can expect to be asked to explain yourself. The same applies if you allow someone else to copy your work. The grader will be watching out for assignments that look suspiciously similar to each other (or to my solutions). Besides which, if you do not do your own assignments, you *will* do badly on the exams, because the struggle to figure things out for yourself is an important part of the learning process.

Some packages that you will probably need too:

```
library(MASS)
library(tidyverse)

## -- Attaching packages ---------------------------------- tidyverse 1.2.1 --
## v ggplot2 3.1.0     v purrr   0.2.5
## v tibble  1.4.2     v dplyr   0.7.8
## v tidyr   0.8.2     v stringr 1.3.1
## v readr   1.3.1     v forcats 0.3.0
## -- Conflicts ------------------------------------- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## x dplyr::select() masks MASS::select()
```

If you're planning to use `select`, you will want the `tidyverse` one rather than the `MASS` one, so load them in this order.

Hand in questions 2 and 4 below. These have a lot of parts, but most of the parts are meant to be short.

1. Make sure you have worked through, or at the very least read through, chapter 16 of PASIAS. Problems to concentrate on here: 16.4, 16.5, 16.6, 16.8, 16.10.

2. The weight of a baby when it is born is an important determinant of how much extra care the baby may need after it is born. A standard definition of a "low birth weight" baby is one that weighs less than 2500g (5 lbs 8 oz). The Wikipedia article on low birth weight is a good starting point.

   What factors are associated with a baby being of low birth weight? A data set was collected of 189 women, for each of whom 11 variables were measured. This data set is stored as `lowbwt` in package `aplore3`.

   (a) (2 marks) To obtain the data set, first install the package (with `install.packages`) and then load it (with `library`). Display some of the data set, which is called `lowbwt`.

   (b) (2 marks) Each line of the data file represents one woman. How do you know that a logistic regression will predict the probability of a low birth weight, rather than a non-low birth weight? Explain briefly. (This is the column `low` in the data set.)

(c) (2 marks) Use logistic regression to predict the probability of a low birth weight as it depends on the mother's age, weight at last menstrual period, race, and number of first trimester physician visits. Display the results.

Notes: (i) you will probably have to consult the help for this data set to find out which variables these are. To do this, make a new code chunk, and in it put a question mark followed by the name of the data frame. Running this should display the help for the data in the Help pane bottom right. View - Panes - Show All Panes (Control-shift-0) will display your notebook and the help, if you are now looking only at the help. (ii) The number of physician visits is categorized as none, one, two or more (labelled as "two, etc"). That's OK. Use it as is.

(d) (2 marks) We are going to remove explanatory variables that do not add anything significant to the model. Explain briefly why we should use `drop1` rather than `summary` to assess this.

(e) (3 marks) Use backward elimination to build a better model for low birth weight. Do this using `step` or by hand using `drop1` and `update`, whichever you prefer. Display the output for your final model. Use $\alpha = 0.10$. You should find that `race` and `lwt` are the two explanatory variables that remain.

(f) (3 marks) From the output of your final model, describe the effects of each remaining explanatory variable on the probability of the baby being of low birth weight.

(g) (2 marks) Find the quartiles of `lwt`, and obtain a list of the different races. Save these in variables for the next part (where we do some predictions).

(h) (3 marks) Make a data frame containing all combinations of the weight at last menstrual period and race whose values you obtained in the previous part. Predict the probability of a low-birth-weight baby for each of these, and display the results side by side with the values they are predictions for.

(i) (2 marks) Explain briefly how the predicted probabilities for each combination of `lwt` and `race` are consistent with your assessment of the slopes in part (f).

3. Work through, or look at the problems in Chapter 17 of PASIAS that relate to ordinal responses: 17.1, 17.2, 17.8, 17.10. You don't need to worry about nominal responses yet (that's on assignment 3).

4. What chemical qualities does a good wine have? 4898 varieties of white wine from a particular area of Portugal were tested in the lab, obtaining concentrations of various chemicals. A team[3] of wine experts also rated the overall quality of each wine. The ratings were originally on a scale of 0–10, but very few of the extreme ratings were given, so I have collapsed the rating scale to one of Excellent, Good, Moderate and Poor. This is in a (text) column called `qual`. The data are in a CSV file at `http://www.utsc.utoronto.ca/~butler/d29/wine_quality.csv`. Can we predict on the basis of the chemical measurements how highly the wine will be rated?

(a) (2 marks) Read in the data and verify that you have the right number of rows and more columns than will fit on the page.

(b) (2 marks) Why would an ordinal logistic regression be better than a nominal (multinomial) one for these data? Explain briefly.

(c) (2 marks) Create a suitable ordered response factor, or explain briefly why you can turn `qual` into a factor directly.

(d) (2 marks) Fit an ordered logistic regression predicting quality from volatile acidity and residual sugar. To enable the grader to check that you have done the right thing, "print" the results: that is, type the name of your fitted model object on a line by itself and include the output from this in what you hand in. (We will investigate the results further later on.)

(e) (2 marks) Can either one of these two explanatory variables be dropped from the model? Obtain some output that will help you decide.

(f) (4 marks) The quartiles of `volatile_acidity` are 0.21 and 0.32, and of `residual_sugar` are 1.7 and 9.9. Make a data frame containing all combinations of these values, and use it to predict that wines with those values of `volatile_acidity` and `residual_sugar` will be awarded each of the ratings.

(g) (2 marks) Describe the effect of increasing residual sugar on the wine quality rating.

(h) (2 marks) Describe the effect of increasing volatile acidity on the wine quality rating.

Notes

[1] Does this surprise you? It surprised me, because I'm used to a higher weight being a risk for health problems.

[2] This may bother you, but it's what the data say. It's now up to the people who are analyzing the data for real to rule out other possible reasons for the difference, before you get to say that there is racism at play here.

[3] In case you thought it was one person rating 4898 wines.

[4] This means that if you reverse the order of things in the group-by, you'll get proportions of wines that were in each rating category for the correctly-identified ones and the wrongly-identified ones. I got this wrong the first time. Try it yourself and see what adds up to 1.